INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

Department of Computer Science and Engineering

COURSE SYLLABUS

Course Information Details

Course Information	Details
Semester	3rd Semester
ISCED Code	0613
Course Code	CSE-2340
Course Title	Software Development 1
Credit Hours	2
Contact Hours	4 laboratory classes per week (50 minutes each; 200 minutes total)
Type	Core, Engineering
Prerequisite	CSE-1222 (Computer Programming 2 Lab)
Co-requisite	CSE-1221 (Computer Programming 2)
Delivery	Laboratory / Practical
Course Structure	15 teaching weeks × 4 classes; two 100-minute studio blocks per week; separate 100-minute mid-term practical slot

Course Assessments

Assessment Component	Marks	Assessment Format
Attendance	10	Continuous
Mid-term In-lab Practical	20	Individual, 100-minute practical assessment
Six Mini Projects / Continuous Assessment	40	Individual projects with dedicated evaluation classes
Final Group Project / Showcase	30	Team project, individual ownership verification and viva
Total	100	

Course Rationale

This course provides a practical foundation in modern front-end development through an AI-assisted, student-driven software development studio. Students progress from web architecture and semantic HTML through CSS, responsive design, JavaScript, DOM manipulation, asynchronous APIs, React, routing, ShadCN/ui, Firebase Authentication, Firestore, Git/GitHub workflows and deployment. Students may use AI tools as development partners to analyze requirements, explore implementation options, generate and review code, and investigate errors. The course emphasizes independent verification, debugging, technical judgment and individual ownership rather than line-by-line code reproduction. The existing technical scope and progressive project structure are retained.

Objectives

No.	Course Objective
1	Apply modern front-end development principles and practices to build practical web interfaces and applications.
2	Analyze interface structure, responsive design decisions, accessibility, client-side behavior and implementation choices.
3	Use HTML, CSS, modern JavaScript, DOM APIs, asynchronous APIs, React, routing, component libraries, authentication, persistence, Git/GitHub and deployment tools.
4	Develop and integrate a responsive interactive front-end application using AI-assisted development workflows while independently verifying correctness and quality.
5	Use AI tools responsibly to analyze requirements, evaluate generated implementations, debug software and explain technical decisions.
6	Work effectively as an individual and team member, communicate technical work, review implementations and demonstrate ownership of submitted code.

Mapping of CLO–PLO–DL–KP–EP–EA

CLO	PLO	DL	KP	EP	EA	Teaching–Learning Strategy	Assessment Strategy
CLO1	PO1	C3	K3	P1	–	AI-assisted problem analysis, guided discovery,	Mini projects, class performance and mid-term

						laboratory challenges and supervised practice.	practical.
CLO2	PO3	C4	K3	P1	–	Requirements analysis, DevTools inspection, testing, debugging and studio critique.	Mini projects, practical examination and project review.
CLO3	PO5	C3	K5, K6	P1	–	AI-assisted implementation, supervised build/debug, code inspection and practical lab work.	Mini projects, practical evaluation and individual modification.
CLO4	PO5	C6	K5, K6	P1	–	Integration studio, iterative development, testing and guided project development.	MP4–MP6 and final group project.
CLO5	PO10	C5/A3	K5	P2	–	AI-use reflection, peer review, code review, demonstration and individual viva.	Project evaluation, AI-assisted development log, PR review and viva.

Revision principle: The course asks whether a third-semester student can independently build, inspect, debug and improve a front-end application in an AI-enabled development environment. Core foundations remain; delivery shifts to challenge-driven studios.

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Revised Weekly Course Content

Each week contains four 50-minute laboratory classes, arranged as two 100-minute studio blocks across two days. Day 1 (C1–C2) emphasizes problem discovery, planning and implementation. Day 2 (C3–C4) emphasizes extension, debugging, verification and demonstration.

Week	Day 1 — C1 + C2 (100 minutes)	Day 2 — C3 + C4 (100 minutes)
1	C1: Web architecture: client/server, DNS, HTTP request/response, status codes, browser rendering, hosting; development environment: VS Code, Node/npm, Live Server, first HTML file. C2: Set up the development environment; create, run and troubleshoot the first HTML file.	C3: Semantic HTML: headings, text, links, images, forms and document structure; Git basics: init, add, commit, push, repository and .gitignore. C4: Apply Git basics to the Week 1 project; repository, commit, push and .gitignore workflow.
2	C1: Navigation, forms and tables; accessibility essentials: landmarks, alt text, labels, heading order, keyboard navigation, focus and contrast. C2: Build navigation, forms and tables; perform an accessibility correction pass.	C3: CSS application methods, selectors, text properties and units; box model, margin, padding, width/height, display and backgrounds. C4: Build and debug layouts using the box model, display and backgrounds.
3	C1: Flexbox, CSS Grid and responsive/mobile-first design. C2: Build responsive Flexbox/Grid layouts and inspect them in DevTools.	C3: Specificity, cascade, pseudo-classes and practical DevTools inspection; build a complete static page. C4: MP1 evaluation: semantic HTML/CSS responsive page and deployment.
4	C1: Design handoff using Figma or Pixso: frames, spacing, measurements and asset export; CSS positioning: relative, absolute, fixed, sticky and z-index. C2: Read a prepared design handoff and implement positioning requirements.	C3: Tailwind CSS v4: setup, typography, spacing, responsive flex/grid; build a responsive Tailwind page. C4: Reproduce and modify a responsive Tailwind page.
5	C1: Responsive deployment, image sourcing and basic image optimization; JavaScript fundamentals: variables, primitives and template literals. C2: AI-assisted JavaScript challenge: build a student performance calculator; analyze requirements, implement calculations, validate inputs and display results.	C3: Arrays, objects and functions through a student/course data-management challenge; inspect, test and modify AI-assisted implementations. C4: MP2 evaluation: own responsive design using Tailwind, deployed. (If already completed, use this slot for individual evaluation/check-off.)
6	C1: DOM selectors, DOM content and attributes, classList, createElement and appendChild through an interactive task-management challenge. C2: Extend the task manager: add, edit, delete and complete tasks; test edge cases and debug.	C3: Events, addEventListener and preventDefault through an interactive quiz or form application. C4: Extend the application with an unexpected requirement; independently modify, test and explain the implementation.
7	C1: Array methods for UI work: map, filter, forEach and find through a searchable/filterable product or student list; midterm preparation. C2: Integrated JavaScript/DOM practical challenge; supervised debugging and individual readiness check.	C3-C4: MID-TERM PRACTICAL EXAMINATION: 100 minutes, individual, 20 marks. HTML, CSS, JavaScript interaction and debugging; no React or Firebase.
8	C1: Modern JavaScript essentials: arrow functions, spread/rest, destructuring, optional chaining, object access, iteration and truthy/falsy; refactor an existing application. C2: Extend and refactor an existing codebase; identify and fix defects, then implement a new feature.	C3: GitHub workflow: repositories, branches, commits and pull requests; team formation and final project brief. C4: MP3 evaluation: DOM application or debug-an-existing-codebase task; individual explanation and live modification.
9	C1: HTTP/HTTPS and APIs; synchronous vs asynchronous JavaScript; Promises, async/await and try/catch/finally through an unfamiliar API challenge.	C3: JSON, fetch, rendering, loading/error/empty states and Network tab; students build a searchable API-driven interface. C4: Final group project proposal review and scope

	C2: Read API documentation, identify endpoints and implement asynchronous requests; handle failures and invalid responses.	approval; SDLC and Agile: user stories, backlog, sprint planning, implementation, testing, review and retrospective.
10	C1: Build an API-driven application with dynamic rendering, loading/error/empty states, search and filtering. C2: Extend and debug API-driven interactions; investigate network failures and unexpected response structures.	C3: localStorage: primitive values, objects and practical cart/saved-item persistence patterns. C4: MP4 evaluation: API-driven application with local persistence and deployment.
11	C1: Vite scaffolding, components and JSX; convert an existing interface into React; Props, conditional rendering and lists. C2: Create components, pass props, render lists and conditional UI; explain component boundaries.	C3: Event handling and useState; useEffect, API calls with state and practical React project structure. C4: Build and debug a small state-driven React application; independently modify a component or state behavior.
12	C1: Immutable state updates and lifting state up; React Router: routes, Outlet, Link and NavLink. C2: Update state safely and implement routes, nested layouts and navigation.	C3: Dynamic routes, useParams, useNavigate, 404/loading states; ShadCN/ui prepared setup and practical Button, Card, Dialog and Table components. C4: Build routed React pages using ShadCN/ui; test navigation and component behavior.
13	C1: Controlled React forms and practical validation; reusable components, props and children. C2: Build controlled forms and reusable components; validate user input and test edge cases.	C3: React debugging: component inspection and common state/prop problems; review code quality and maintainability. C4: MP5 evaluation: React SPA with routing, reusable components and ShadCN/ui, deployed.
14	C1: Firebase project and SDK; Google sign-in/sign-out; email/password registration/login, auth state and private routes (without Context API). C2: Implement authentication flows and private-route behavior; test authenticated and unauthenticated states.	C3: Firestore: one collection, basic read/write; supplied security-rules starter; configuration safety. C4: MP6 evaluation: authentication + one Firestore collection, deployed.
15	C1: Practical project polish: responsive behavior, accessibility checks and lightweight Lighthouse audit; GitHub pull-request review and final submission. C2: Polish, test, audit and prepare the final project submission; verify README, deployment and feature completeness.	C3: Integration studio: final debugging, system testing, peer review and individual ownership verification. C4: FINAL GROUP PROJECT SHOWCASE AND PRESENTATION: team demonstration and individual viva.

Note: Week 5 C1 represents the class already completed. The revised AI-assisted approach begins from Week 5 C2.

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AI-Assisted Development Studio: Standard Weekly Delivery

The four 50-minute classes are grouped into two consecutive 100-minute sessions. The instructor acts as facilitator, technical mentor, reviewer and evaluator rather than providing line-by-line code demonstrations.

Day 1 — Problem Discovery and Implementation

Time	Activity
0–10 min	Introduce a real-world problem, constraints and expected outcome.
10–20 min	Students analyze requirements, identify concepts and plan with an AI assistant.
20–55 min	Independent AI-assisted implementation; instructor circulates and asks diagnostic questions.
55–70 min	Run the application, test expected behavior and identify errors.
70–90 min	Implement an extension or additional requirement.
90–100 min	Commit work to GitHub and submit a brief progress record.

Day 2 — Engineering, Debugging and Verification

Time	Activity
0–10 min	Review previous work and identify unfinished tasks.
10–35 min	Implement additional features or improve the existing solution.
35–55 min	Debug, test, inspect code and verify AI-generated suggestions.
55–75 min	Instructor introduces an unexpected requirement or change request.
75–90 min	Students independently modify the application and explain their decisions.
90–100 min	Individual explanation, GitHub commit and exit assessment.

Instructor's Role

Introduce problems, clarify requirements, monitor progress, ask probing questions, identify misconceptions, review implementation decisions and verify individual understanding.

Student's Role

Analyze tasks, use an AI assistant of choice, plan and implement features, inspect generated code, test behavior, debug errors, document progress, commit changes and explain or modify their own implementations.

Dedicated Mini-Project Evaluation Plan

Project	Due / Evaluation	Deliverable	Marks	Evaluation Focus
MP1	Week 3 C4	Semantic HTML + CSS + responsive layout + deployment	5	Structure, styling, responsiveness, deployment and individual explanation.
MP2	Week 5 C4	Own responsive design + Tailwind + deployment	5	Design translation, Tailwind usage, responsive behavior, repository and deployment.
MP3	Week 8 C4	DOM application or debugging task	7	DOM logic, events, interaction, debugging, ownership and live modification.
MP4	Week 10 C4	API application + localStorage + loading/error states + deployment	7	API handling, UI states, persistence, debugging and deployment.
MP5	Week 13 C4	React SPA + routing + reusable components + ShadCN/ui + deployment	8	Components, props/state, routing, UI library and individual modification.
MP6	Week 14 C4	Firebase Authentication + one Firestore collection + deployment	8	Authentication, private access, persistence, configuration safety and deployment.

Recommended Individual Evaluation Procedure

Step	Procedure
1	Check repository and deployed URL before class.
2	Ask the student to demonstrate the required feature.
3	Ask 1–2 questions about implementation decisions or a selected code segment.

4	Introduce one small change, edge case or debugging task.
5	Observe testing and explanation.
6	Record the rubric result.

Mini-Project Evaluation Rubric

Criterion	Weight	What is Checked
Functional implementation	30%	Required features work correctly.
Technical understanding	20%	Student explains logic, structure and decisions.
Debugging and independent modification	20%	Student investigates errors and implements a change.
Code quality and organization	15%	Structure, naming, readability and maintainability.
AI utilization and verification	10%	AI suggestions are tested and evaluated.
Demonstration and explanation	5%	Student communicates implementation clearly.

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Mid-term Practical Examination: 100 Minutes (C3–C4 Block)

Item	Specification
Duration	100 minutes (C3–C4)
Marks	20
Nature	Individual practical assessment
Permitted scope	Semantic HTML, CSS, responsive layout, JavaScript fundamentals, DOM and meaningful interaction covered before the mid-term.
Suggested requirements	Implement a small application from a problem statement; include input handling, dynamic behavior, validation and debugging.
Assessment emphasis	Implementation, problem-solving, responsiveness, interaction, testing and individual understanding.

AI Policy for the Mid-term

The mid-term measures individual practical competency. The instructor specifies permitted AI-use conditions before the examination. Students must not use AI to bypass individual demonstration or examination rules.

Assessment Strategy

Assessment	Marks	CLO Alignment	Nature
Attendance	10	CLO5	Continuous
Mid-term Practical	20	CLO1, CLO2, CLO3	Individual, 100-minute practical
Mini Project 1	5	CLO1, CLO2, CLO3	Individual + dedicated evaluation
Mini Project 2	5	CLO2, CLO3	Individual + dedicated evaluation
Mini Project 3	7	CLO2, CLO3	Individual + dedicated evaluation
Mini Project 4	7	CLO3, CLO4	Individual + dedicated evaluation
Mini Project 5	8	CLO3, CLO4	Individual + dedicated evaluation
Mini Project 6	8	CLO4	Individual + dedicated evaluation
Final Group Project / Showcase	30	CLO4, CLO5	Team project + individual viva
Total	100		

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Final Group Project

Teams of three are formed in Week 8. Teams may propose their own problem and application idea, subject to instructor approval. The project integrates React, routing, ShadCN/ui, Firebase Authentication, Firestore and deployment.

No.	Minimum Requirement
1	At least three routes.
2	Authentication with protected routes.
3	At least one Firestore collection with basic read/write.
4	At least one meaningful ShadCN/ui feature.
5	Responsive design and deployment.
6	README with setup, features and contribution information.
7	Each member owns at least one feature end-to-end.
8	Each member works on an individual branch and opens a pull request.
9	Each member makes at least eight meaningful commits.
10	Each member completes an individual viva and demonstrates their own contribution.

AI Use in the Final Project

Teams may use AI for planning, code generation, debugging, documentation and review. Each member remains responsible for correctness, security and maintainability, should disclose significant AI assistance, and must explain, test and modify their own code during viva.

Topics Intentionally Retained as Core

Area	Reason
HTML + semantic structure	Foundation for every web interface.
CSS + Flexbox + Grid + responsive design	Essential for front-end layout and usability.
JavaScript + DOM + events	Core bridge from static pages to interactive applications.
Promises + async/await + fetch + UI states	Required for real API-driven front ends.
Vite + React components/props/state/useEffect	Modern application development foundation.
React Router	Required for realistic SPAs and final project.
ShadCN/ui	Retained as a practical subset.
Firebase Authentication and Firestore	Retained at practical scope.
Git/GitHub	Repository, commits, branches and pull requests.
Deployment and six mini projects	Publish working applications and demonstrate progressive competency.

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Course Delivery and Teaching–Learning Strategy

Teaching–Learning Strategy	Description
Strategy	Implementation
Challenge-driven learning	Each studio begins with a realistic problem statement, constraints and expected outcome.
AI-assisted learning	Students may use ChatGPT, Gemini, Claude, Copilot or another preferred AI assistant to analyze, plan, implement and investigate.
Independent implementation	Students build features themselves and remain responsible for the resulting code.
Code inspection and verification	Students inspect generated code, test behavior, check edge cases and verify proposed fixes.
Debugging-based learning	Students diagnose errors and explain causes rather than replacing code blindly.
Unexpected requirement changes	The instructor introduces small change requests to assess adaptability and individual understanding.
GitHub-based evidence	Students commit progress, document work and use branches/pull requests where appropriate.
Peer review and studio critique	Students review implementations and discuss trade-offs, usability and code quality.
Dedicated project evaluation	One complete 50-minute class is reserved for each mini-project check-off.
Final integration studio	Students test, debug, polish and demonstrate the final group project.

AI-Assisted Development Policy

No.	Policy
1	AI tools are permitted during regular laboratory development activities; students may choose their preferred assistant.
2	Students must not blindly accept generated code. They must run, test and verify each implementation.
3	Students are responsible for understanding, explaining, modifying and debugging all submitted work.
4	Significant AI contributions should be acknowledged in the project README or development log.
5	Students must not submit credentials, private keys, personal data or confidential information to AI tools.
6	AI output must be reviewed for correctness, accessibility, security, maintainability and suitability for the task.
7	Use of AI does not replace individual assessment, live demonstration or viva.

AI-Assisted Development Log

For each mini project, students maintain a concise record: problem understanding; AI contribution and representative prompts; implementation decisions; testing and verification; one error investigated and resolved; and personal reflection on what they can now implement independently.

Core Graduate Capability After CSE-2340

A student completing CSE-2340 should be able to design, implement, inspect, debug and deploy a responsive front-end application using HTML/CSS and modern JavaScript; manipulate the DOM; consume asynchronous APIs; build React SPAs with routing and state; use ShadCN/ui; implement Firebase Authentication and basic Firestore persistence; collaborate through Git/GitHub; and use AI-assisted development tools responsibly while independently verifying software quality.

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Dependency with Later Courses

CSE-2340 provides the front-end foundation without consuming the server-side and database depth required by later courses. CSE-3532 Web Programming can therefore emphasize server-side programming, REST design, database integration, validation and project architecture rather than repeating the full front-end sequence.

Recommended Learning Materials

Resource	Use
MDN Web Docs	Primary reference for HTML, CSS, JavaScript, DOM and browser APIs.
web.dev – Learn CSS / Learn Forms	Practical CSS, responsive and form guidance.
javascript.info	JavaScript fundamentals and browser-side programming.
Eloquent JavaScript	Supplementary JavaScript reference.
React Documentation	Components, state, effects and React patterns.
Vite Documentation	Modern React development environment.
React Router Documentation	Routing and navigation.
Tailwind CSS Documentation	Utility-first responsive styling.
ShadCN/ui Documentation	Practical component usage.
Firebase Documentation	Authentication, Firestore and deployment.
Git / GitHub Documentation	Repositories, commits, branches and pull requests.
Pro Git, 2nd Edition	Supplementary Git reference.
Figma / Pixso Documentation	Basic design handoff and asset inspection.

Assessment Notes

No.	Assessment Note
1	The six mini projects are individual and progressive.
2	Each mini project is preceded by relevant challenge-based practice.
3	Each mini project has a dedicated evaluation class.
4	Evaluation may use a deployed URL, repository link, rubric, short demonstration and one small live modification.
5	AI tools may be used openly during regular development; students must verify, explain and modify submitted work.
6	The AI-assisted development log records problem understanding, AI contribution, verification, debugging and reflection.
7	The final group project is team-based, but individual ownership is verified through features, branches, pull requests, commits and viva.
8	The mid-term is a 100-minute individual practical examination occupying the Week 7 C3–C4 block; AI-use conditions are announced before the examination.

Revision Summary

The syllabus retains the existing course identity, credit, contact hours, technical scope, assessment marks, six mini-project structure, mid-term practical and final group project. The principal revision is the delivery model: four 50-minute classes are organized into two 100-minute AI-assisted development studios each week. Students learn through realistic challenges, independent implementation, verification, debugging, change requests and demonstration. The instructor facilitates and evaluates learning rather than teaching code line by line.